

EMEM: A Content-Addressed, Verifiable Earth-Memory Protocol for AI Agents over Foundation-Model Embeddings

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paper doi:10.5281/zenodo.20706893 · code github.com/Vortx-AI/emem (Apache-2.0) · live emem.dev · method: whitepaper v3, emem @18adb67



THE PROBLEM

Agents that work together drift apart.

Each agent reads the world through its own searches and summaries. After a few handoffs two agents no longer agree on where a place is, what was measured or when, and neither can show the other which number is right. Agreement does not rescue them (R2).

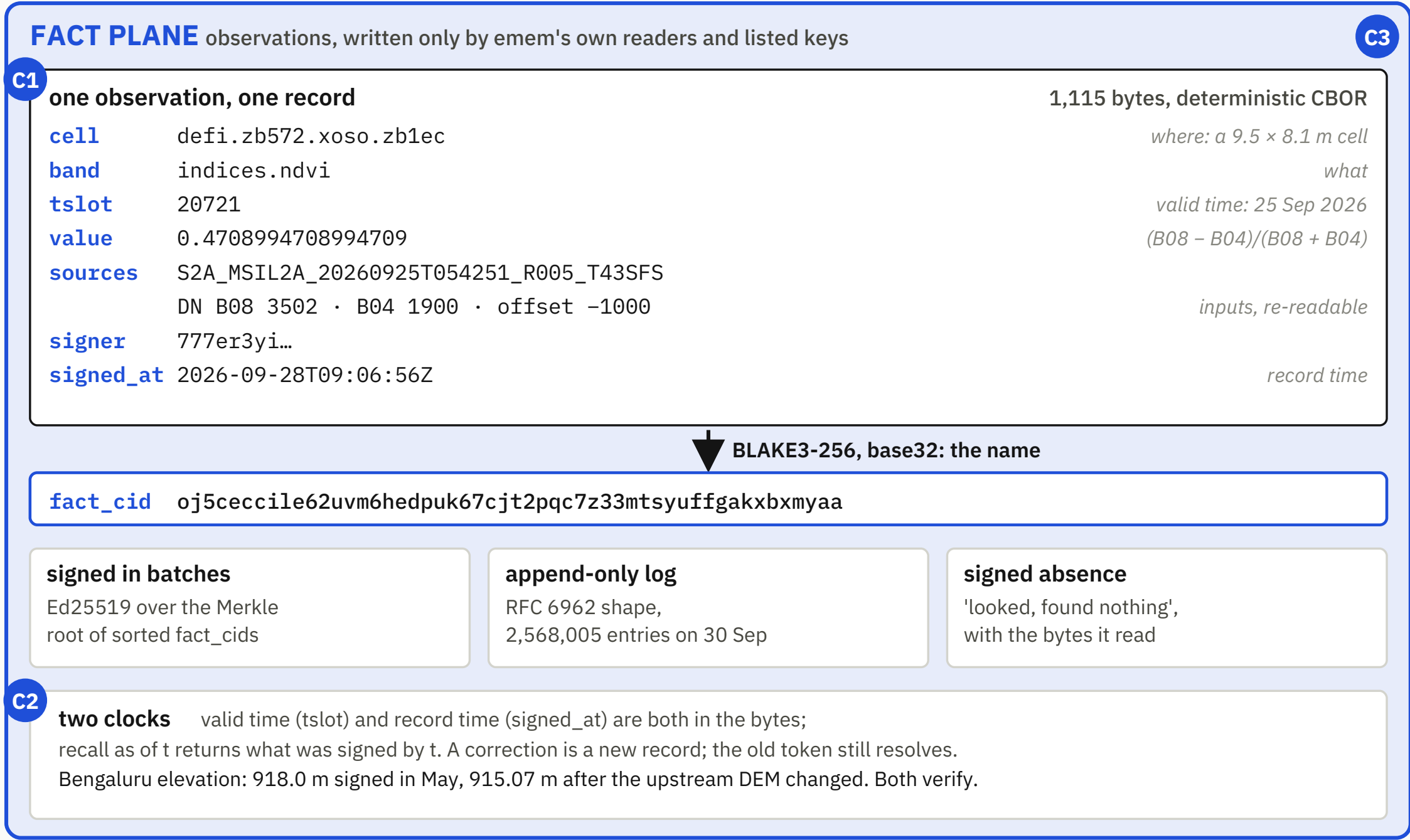
THE WORLD WRITES



a 2.02 GB scene; the square marks the cell of the NDVI record (drawn 100 m wide)

OTHER WRITERS

Copernicus DEM · Overture Maps · weather · Earth embeddings
enrolled devices (next)



Keyed by where, what and when; named by the hash of its bytes.

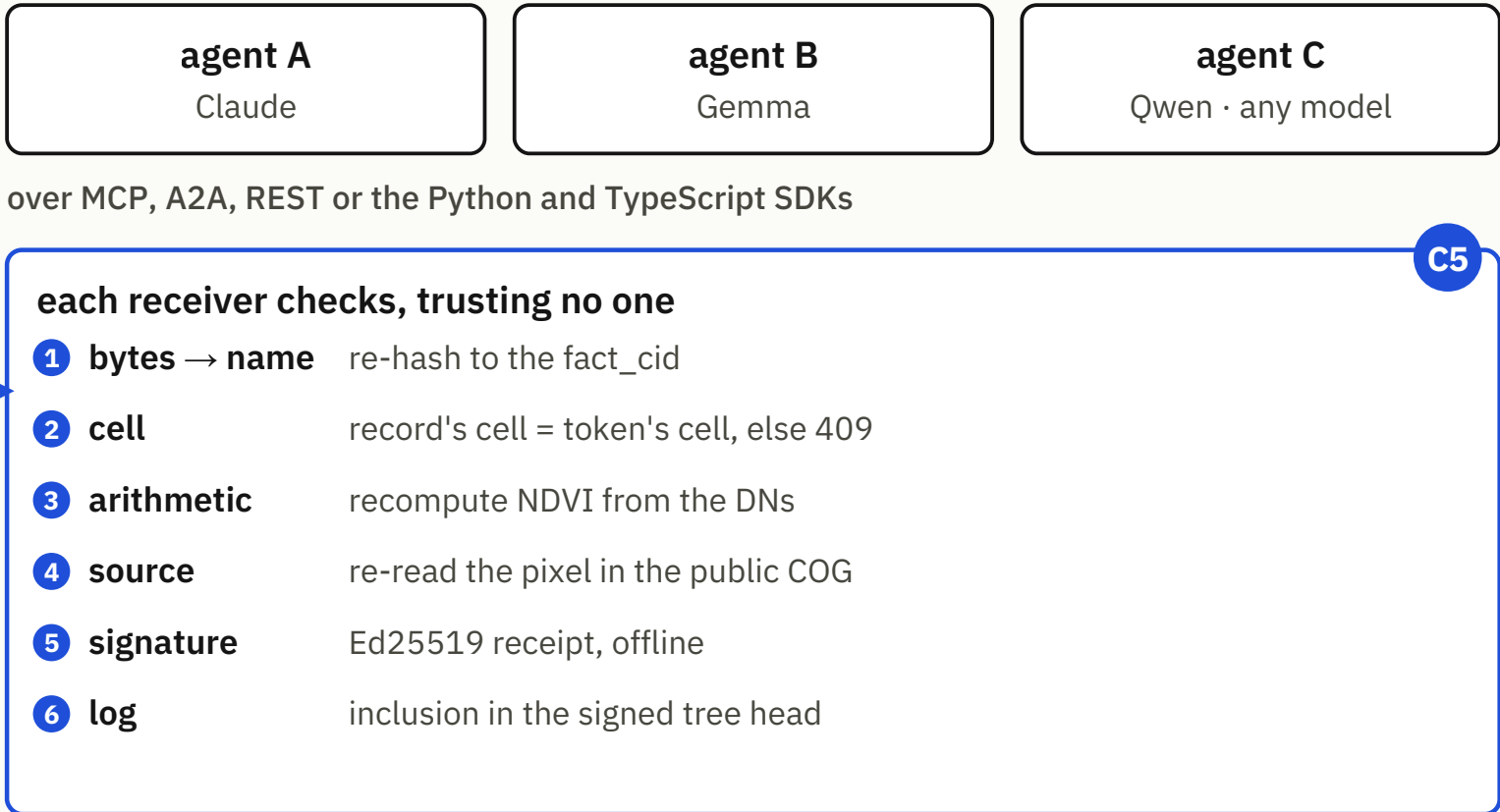
Only the 84-character name travels; the bytes stay where anyone can re-read them.

THE INVENTION

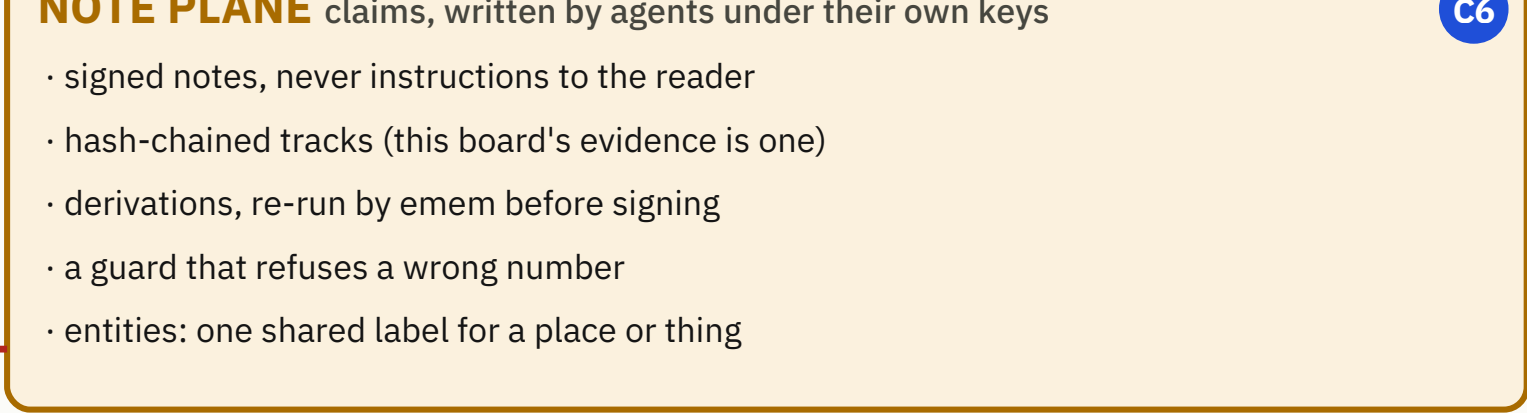
One place, one address. One observation, one signed record. The record's name, not a paraphrase, is what crosses between agents.

A memory of the physical world that agents share and no agent controls: **written by machines, named by the hash** of its bytes, **kept on two clocks**. In a 17-case mutation test on real bytes, prose passed all 15 corruptions it can carry; the full check chain passed none (R1).

AGENTS READ, CHECK AND CITE



over MCP, A2A, REST or the Python and TypeScript SDKs



Every check runs with stock libraries: Python stdlib, blake3, cbor2, pynacl.

Six contributions

each a mechanism in emem's code at 18adb67, each with a measurement on the live system

C1 An address for every observation

A value is keyed by where, what and when, and named by the hash of its exact bytes. The key finds a record; the name proves which one.

```
k = (cell64, band, tslot)
fact_cid = base32(BLAKE3-256(CBOR(f)))
token = emem:fact:<cell>:<fact_cid>
```

A **1,115-byte** record read from a **2.02 GB** scene travels as **84** characters §4; ten client paths re-derive the same fact_cid.

C4 One token algebra, pixel to archive

The same naming covers a value, a field, a field through time, a set of fields, any file hashed where it lies, and one handle for many facts. Each family declares how strongly it binds.

```
fact → raster → cube → rasterset
tree = Merkle(chunks of a file, in place)
bundle = one handle over ≤ 256 fact_cids
```

A 443 × 453 px B04 field under one token §5; a 5-scene cube whose members name the scene actually used §6; a **238.7 MB** COG hashed in 292 chunks without moving it §2; 8 records behind one 38-character line §16.

C2 A memory on two clocks

Valid time (when the Earth was seen) and record time (when it was signed) are both in the hashed bytes. New evidence appends; nothing is rewritten.

```
as of t: argmax_signed_at ≤ t (signed_at, cid)
receipt binds (as_of_tslot, as_of_signed_at)
log leaf = BLAKE3(0x00 || BLAKE3(CBOR(attestation)))
```

Bengaluru elevation moved **918.0** → **915.07 m** when the upstream DEM changed §12 §13. Every read takes both time bounds, and the receipt signs them (R4).

C5 Verification that needs no trust

A receiver checks bytes, place, arithmetic, source pixel, signature and log with stock libraries. Each record shows how deep it can be checked.

```
accept ⇔ BLAKE3(b) = cid ∧ cell(b) = cell(token) ∧ NDVI(DN(b)) = value ∧ COG(row, col) = DN(b) ∧ Ed25519(σ) ∧ leaf ∈ STH
depth: recomputable > reunnable > trace-verified > checkpoint-signed > attester-only
```

Our script: **15 links, 17 checks, 17.9 s**, no emem code §22. An unaffiliated agent's clean-room verifier from /v1/verifier_spec: **725** requests, zero errors, five tampered receipts rejected.

C3 Machines write facts; agents write claims

Only emem's readers of registered archives, enrolled devices and operator-listed keys write the fact plane. Agents sign notes under their own keys: data, never facts, never instructions to the reader. When nothing qualifies, emem signs an absence.

```
cell(bytes) ≠ cell(token) → refused (HTTP 409)
absence = (cell, band, tslot, reason_cid, sources)
```

Venice: *no carriageway within 50 m* in Overture 2026-09-23.1, with the row groups it read (exhq6lps...). /v1/plane/conformance tests that no tool takes a caller's value.

C6 Computation others can recompute

An agent derives a value over signed facts and emem re-runs the pure operation before recording it. A guard refuses a sentence whose number disagrees with the fact it cites.

```
derive accepted ⇔ ulp_gap(v_caller, v_emem) = 0 (≤ 4 for n-ary sum, mean)
guard: stated ≠ signed at the written precision → deny
```

Our NDVI change of **-0.1298** (25 Sep 2025 → 2026) was recomputed bit-identical and signed under our key §11. The guard refuses "35 °C" citing a signed 28.0 °C §14.

What agents do with it

one memory for every model, running at emem.dev: 2.57 M signed log entries, 168 versioned algorithms, 46 source schemes (30 Sep 2026)

Research a place, no web ask · 168 recipes	Hand work to another agent emem:bundle	Keep long work alive emem_memory_create	Agree on what a thing is emem_entity_resolve	Explain why a number moved emem_change_attribution	Catch a contradiction emem_memory_contradictions	Turn documents into evidence emem:tree	Compute for recomputation emem_derive
Flood, heat, crop or solar scores from signed readings; each recipe cited by id.	Exact signed bytes behind one line, on any model or vendor.	Signed notes outlast sessions, compaction and restarts. Notes are public.	A farm or a building gets one identity; different phrasings converge on it.	Names the terms behind a change, each with its fact ids.	Finds records that disagree; the guard refuses a wrong number.	Lab reports and land records become signed units under one token.	Pure operations are re-run by emem before the result is recorded.
One token, three vendors Claude (through MCP), Gemma 3 on Amazon Bedrock and Qwen 2.5 running locally all end with the same fact_cid and value. The decoding is emem's, not the model's: two of the three were handed the same resolved response.	A public dispute, settled by bytes geo.qa's agent re-derived a Doha road fact and reported 9.8 m against emem's 5.4 m. Re-measuring from the full-precision coordinate inside the fact gave 5.4 m, and that agent withdrew its number in a signed note.			Regulation an auditor can re-check eudr.dev checks farm plots against the EU Deforestation Regulation cut-off with emem's forest facts, and prepares Annex II statements that an auditor can re-verify.		Any node, the same receipts The hosted node runs the Apache-2.0 binary, and a receipt minted on one node verifies on another. geo.qa runs a second node; an air-gapped build runs with no network at all.	

Four results

scripts, data and pre-registration hashes: github.com/Vortx-AI/esa_poster/research/repro

R1 · THE MUTATION TEST: 17 CASES × 9 DEPTHS, ON REAL BYTES

Each check stops a failure that the others let through

	prose	JSON	opaque id	hash	binding	signature	log	recompute	re-read
G0 PARAPHRASE, RELAY									
M1 stated value +1 ULP									
M2 stated value rounded to 0.47									
M3 1 ULP changed in served bytes									
M4 MISSENDING									
M5 record cited for another cell									
M6 older record passed as current									
M7 record for another band									
M8 token miscopied by 1 char									
M9 FORGERY, NO KEY									
M10 value forged to 0.45, re-hashed									
M11 cell forged, re-hashed									
M12 date forged, re-hashed									
M13 source scene forged, re-hashed									
M14 offset forged, value recomputed									
M15 THE TRUSTED SIGNER ERRS									
M16 signer: value disagrees with DN's									
M17 signer: DN's of the pixel 10 m south									
M18 signer: second version, not logged									
M19 same bytes, a different entity									
M20 acted on corrupted evidence									

B receives the Keylong record §4 after one corruption; a deterministic verifier, no model in the loop, decides whether B acts on it. The rule "irrigate iff NDVI ≤ 0.4705" flips in 6 of 16 cases. T2 rows use a test key the verifier trusts. Leave-one-out: dropping binding, signature, log, recompute or re-read each lets a case through; the signature subsumes the hash. M15 is the one error emem's own reader made, before 28 Sep §7; only the source re-read catches it, which is how it was found and fixed §8. Units, checkpoints and embeddings are not covered.

Prose and JSON pass all 15 corruptions they can carry. Full emem passes none. Which entity A meant (M17) is out of scope by design.

Discussion

design choices, and where emem stops

Integrity is not truth, so truth is re-read.

A signature fixes who served which bytes, and when. Whether the value is right is tested against the public source each record names. In R1 it is the only check that catches a signer's own error, and it is how the one such error in production was found; no signed byte was rewritten, so every affected record stays datable.

emem sits under guardrails, not beside them.

GeoGuard decides whether a claim should stand; provenance-first composition traces a pipeline; STAC, C2PA and PROV locate files, sign media or describe lineage. emem names the exact observation an agent cites, so any of them can check it: a guardrail can take emem tokens as evidence.

Identity first, compression second.

One token costs about 6× the tokenizer tokens of the bare number (46 vs 8); it buys identity, not brevity. The bundle is the compression: up to 256 facts behind one 38-character line.

Conclusion

1 Agents can hand each other the name of a signed observation instead of a sentence about it; any receiver checks it offline in under 2 ms.

2 Five of the six checks each stop a corruption the others miss. Together they stop all 16 in-scope cases, including the signer's own errors.

3 Records are never rewritten, so the memory answers "what was known, when" across upstream changes and retired encoders.

Encoders change; the memory keeps what they said.

Embeddings enter emem as signed model outputs. Clay, Prithvi-EO-2.0, Galileo and a JEPA head left the code after acceptance §23, and new TESSERA vectors are no longer minted; every vector signed before still resolves and verifies §24. Swapping an encoder never rewrites history.

Retrieval finds text; a reference names bytes.

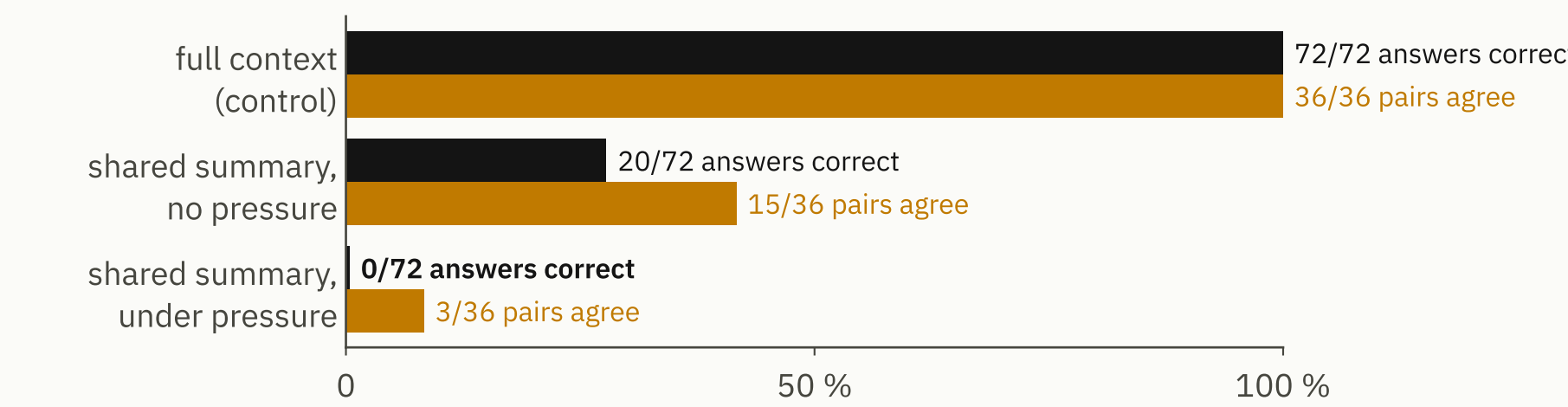
BM25 matched the exact value when the string was in the corpus (16/16). Dense retrieval returned 4 of 142 exact and up to 138 confident misses, a median 252 m off §25. Neither lets the receiver check what it got.

Independent operation is the next step.

Every check runs with stock libraries and a published key, with no call to us, and an unaffiliated clean-room verifier reproduced them (C5). The log is cross-signed with a second node, geo.qa, which Vortx AI also runs. Next: independent operators, provider-signed upstream identity and a multi-vendor agent study. R2 and R3 are pre-registered samples; R1 is deterministic.

R2 · COMPACTION AND HANDOFF

Agreement is not evidence



Pre-registered; Gemma-4-12B and Qwen2.5-7B on one host §17. Under pressure, 3 of 36 pairs agreed while 0 of 72 answers were right (Fisher p = 0.035; pairs share trials, so descriptive). In a separate handoff study, n = 20 per arm §18, the exact value survived prose 2/20, dense retrieval 8/20, BM25 20/20, the emem bundle 20/20.

Agreement between agents cannot certify a value.

R3 · TWO LLMS, EMEM THE ONLY TOOL

A token helps only when checked

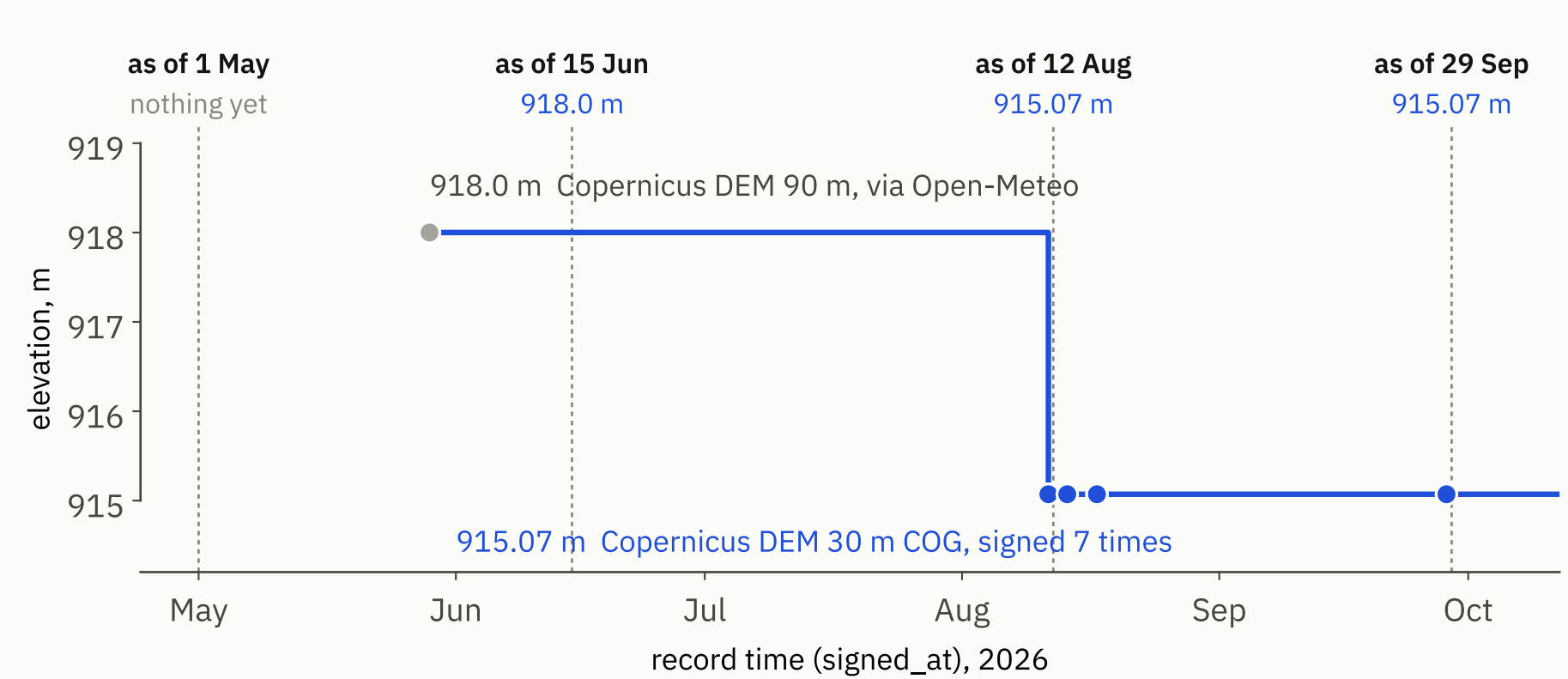
B RECEIVES	B RIGHT	B CHECKED
A's token	10/10	resolved; harness re-hash, receipt 10/10
A's prose, 16 digits	10/10	nothing to check
prose rounded to "0.47"	0/5	nothing to check
token with a forged cell	5/5 declined	MCP error; REST 409

A = Claude Sonnet 5.5, B = Claude Haiku 4.5, pre-registered §22; token vs full prose tied (p = 1); rounded arm exploratory.

The check belongs in the resolver, not in the model's goodwill.

R4 · MEMORY THROUGH TIME

The source changed; both answers still verify



One Bengaluru cell, band copdem30m. elevation_mean §12 §13. In May emem signed 918.0 m from the 90 m Copernicus DEM served by Open-Meteo; from 11 Aug it signed 915.07 m read from the 30 m COG. Nothing was overwritten. Asked "as of 15 Jun", emem still answers 918.0, and every answer re-hashes to its fact_cid (live replay, verify_bitemporal.py). The contradiction detector labels the pair a provider change by one attester, not a dispute between witnesses. Each re-signing mints a new fact_cid, because signed_at is part of the bytes.

An agent that cited the May value can still prove what it saw, and why today's answer differs.

What a verified token guarantees

the exact bytes	guaranteed	hash, signature, log (R1 D, F, G)
the place and time	guaranteed	cell and tslot in the bytes (R1 E)
the derivation	recomputable	signed inputs, named pixel (R1 H, I)
the upstream file	named	COG and pixel recorded; provider signature next
the value's truth	tested	by re-reading the source, not by the signature
the physical entity	out of scope	entity tokens are shared labels (R1 M17)
the decision	out of scope	the agent's, or a guardrail's

TRY IT IN ONE MINUTE

claude mcp add --transport http emem https://emem.dev/mcp

Also in ChatGPT, VS Code, Dify and the GitHub MCP Registry · pip install ememdev · npm i @vortxai/emem · self-host: docker run ghcr.io/vortx-ai/emem. Scan to run R1 yourself: python research/repro/v11/mutation_suite.py, offline, under a second.

EVERY \$ IS SIGNED EVIDENCE

Each \$ is step n of one signed emem track, njedkg1t/7n7qogw...: 28 steps, all 28 verified in ememdemo on 30 Sep 2026. Scan to re-hash every step and its chain in your browser.